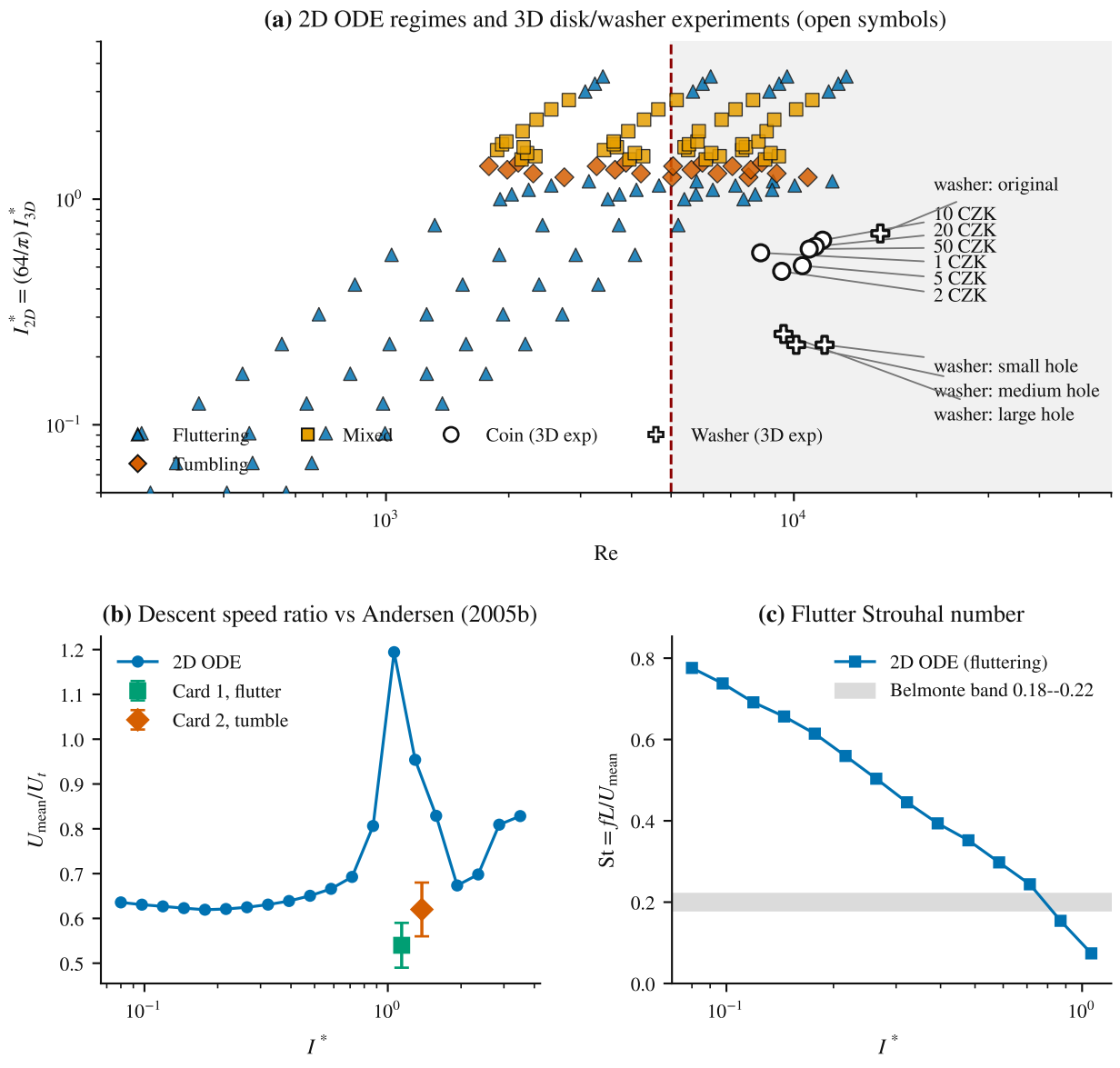
(a) 2D ODE regimes and 3D disk/washer experiments (open symbols)
$I^*_{2D} = (64/\pi)\, I^*_{3D}$
10^0
10^{-1}
10^3
10^4
Re
Fluttering
Mixed
Tumbling
Coin (3D exp)
Washer (3D exp)
washer: original
10 CZK
20 CZK
50 CZK
1 CZK
5 CZK
2 CZK
washer: small hole
washer: medium hole
washer: large hole
(b) Descent speed ratio vs Andersen (2005b)
2D ODE
Card 1, flutter
Card 2, tumble
U_{mean}/U_t
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
10^{-1}
10^0
I^*
(c) Flutter Strouhal number
2D ODE (fluttering)
Belmonte band 0.18--0.22
$\mathrm{St} = fL/U_{\mathrm{mean}}$
0.8
0.6
0.4
0.2
0.0
10^{-1}
10^0
I^*